# João André **Roque Costa**

MSc Graduate, Computer Science and Engineering

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Computer Science Developer with a Master's degree in Computer Science (AI and Data Science) from **Instituto Superior Técnico**.
Experienced in **Algorithm Optimization** and **Solution Design**.
Proficient in **Python**, **C/C++**, and databases, with a strong background in **Machine Learning**, **Deep Learning**.

## Professional Experience

Deloitte — *Coimbra, Portugal*
**Tech Consultant** — *Apr. 2026 - Oct. 2026*

- **Planned** and **developed** a multi-agent application for data migration and generation tasks (Claude SDK, Azure Blob Storage and MongoDB).
- **Ported** and updated an Azure Resource Group between two subscriptions (Azure Managed Instances, Data Factory, Databricks).
- **Automated** an HR report generation process using Python and Playwright.
- **Drafted** a project proposal to a client to automate the documentation processes (RAG, OCR).

Instituto Superior Técnico — *Lisbon, Portugal*
**Invited Teaching Assistant** — *Oct. 2024 - Jul. 2025*

- **Lectured** and **evaluated** practical classes on Databases and Data Science.
- **Assisted** students with hands-on projects in data treatment, visualization, and analysis techniques using Python (pandas/matplotlib).

## Education

Instituto Superior Técnico — *Lisbon, Portugal*
**M.Sc. Computer Science and Engineering** — *Sep 2023 - Nov 2025*

- Specialized in **Artificial Intelligence** and emphasis in **Data Science**, connecting the theory study with practical applications.
- **Relevant Coursework**: Optimization Algorithms, Deep Learning, Cloud Computing and Virtualization.
- Thesis: **Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)**.
  - Developed a novel extension to the MVU algorithm that enables **nonlinear dimensionality reduction** on disconnected datasets.
  - Engineered a decomposition-based approach that **allows for parallel processing**, significantly **improving scalability** for large-scale datasets that were **previously computationally infeasible**.
  - Achieved equal-quality embeddings and preservation of local geometric structures compared to the standard method.

Instituto Superior Técnico — *Lisbon, Portugal*
**B.Sc. Computer Science and Engineering** — *Sep 2020 - Jul 2023*

- **Grew** a strong foundation in Analysis and Synthesis of Algorithms, Distributed Systems, Operating Systems, and Database Theory.
- **Collaborated** on several **full-stack** and **systems-level** projects, emphasizing **memory management** in **C/C++** and relational schema design in **SQL**.

## Extracurricular

### Personal Software Projects (Self-Hosted)

- Engineered an automated betting bot for parimutuel systems using a modified Kelly Criterion for **risk management** and **optimal stake sizing**.
- Developed a custom Discord-based ELO ranking and **automatic matchmaking system**.
- Architected a multi-platform **arbitrage engine** and inventory manager. With automated price monitoring and trade execution across multiple marketplaces.
- Built a full-stack system for **managing multiple modular projects** in a linux nuc (per machine resource management, product price tracking).

## Skills & Competencies

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| **Programming** | **C**, **C++**, **Java**, **Python** (Claude-Agent-SDK, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Playright) |
| **Databases** | **SQL**, **NoSQL**, Microsoft SQL Server (SSMS), PostgreSQL, MongoDB, MySQL, pgAdmin |
| **Data Engineering** | Databricks, **Pentaho PDI**, SQL Profiler, Tuning Advisor (DTA), DataCleaner tool |
| **Business Intelligence** | Pentaho (Server, Schema Workbench, Report Designer), Saiku Analytics |
| **Cloud & Architecture** | **AWS**, **Azure**, ArchiMate, Camunda Modeler, Enterprise Architecture (Sparx) |
| **Developer Tools** | Git, Linux, Bash Scripting, RESTful APIs, LaTeX |
| **Languages** | Portuguese (Native), English (Fluent) |